



SMART HOSPITAL AUTOMATION SYSTEM USING-IOT

A MAJOR PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

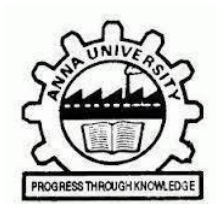
ELECTRONICS AND COMMUNICATION ENGINEERING

ADHIYAMAAN COLLEGE OF ENGINEERING [AUTONOMOUS]

Dr. M.G.R NAGAR, HOSUR - 635 109, TAMILNADU

ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this project report titled “**SMART HOSPITAL AUTOMATION SYSTEM USING-IOT**” is the Bonafide work of **LOGESH BABU R, MATHAN S, MOHAMMED IRFAN S, SRI HARI S** who carried out the research under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

The system describes the best outcome of the ongoing INTER-Health channel coming out from the application of the strategy proposed and developed within the framework of the embedded cum IoT. The narrative IoT based healthcare platform, supporting decentralized and mobile monitoring of assisted livings in highly heterogeneous contexts, has been obtained by integrating other existing heterogeneous, non-interoperable IoT platforms following the INTER-IoT approach to provide value added assisted living mobile healthcare services (respect to the standard “manual” monitoring performed by conventional healthcare centers) in terms of faster detection and

correction of wrong lifestyles or high risk critical situations. All the interoperability aspects and technological choices have been presented to discuss the ease of such integration process, also making easier and faster the interaction between doctor and patient drastically reducing the dropout rate of people leaving the pilot. The device of the system is developed as a portable inlay that can be placed in small places like inside pocket of a jacket. It combines multiple sensors, both of which work in a contact way allowing unobtrusive monitoring. The use of the sensor method is intended for heart pulse, blood rate and body temperature measurement. The sensors emit light of a specific wavelength into the tissue region under investigation, measure the amount of light that passes through the tissue, and arrive at a measurement unit. Since the light intensity at the measurement unit depends on blood content in the tissue, this sensor technique is well suited for cardiac pulse detection except temperature measuring system. The DS18B20 sensor is used to measure the temperature of the human system. The experiment of the system is analyzed by using hardware model.

Keywords: Embedded system, IoT, sensor network etc.,

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LIST OF ABBREVIATIONS

AC	Alternating Current
BP	Blood pressure
BT	Blockchain technology
DApp	Decentralized Applications
DC	Direct Current
GSM	Global system for mobile communication
IC	Integrated Circuit
ILS	Indoor Localization Systems
IOT	Internet of Things
NTC	Negative Temperature Coefficient
WiFi	Wireless Fidelity
WOT	Web of things

CHAPTER 1

INTRODUCTION

In today's scenario, health care problems are increasing at a very high pace like coronary heart diseases, blood pressure and abnormal body temperature etc., causing death rate of 7.2 million people per year hence it is the need of the hour to overcome from all such problems. One of the most attractive application environments in the IoT arena is healthcare. IoT technologies have the potential to boost up many medical applications, such as remote and mobile health monitoring, human activity recognition, reduction of hospitalizations of chronic patients and elderly care at home. The development of new applications and services can help to decrease healthcare costs, improve the quality of life and enhance the patients' experience; however, specific healthcare requirements were often faced developing particular and custom systems that were not interoperable and/or reusable in different, though similar, contexts.

Today there is a surplus of IoT systems that are not enough scalable neither sustainable because they are not able to interoperate and whose integration process is almost expensive as the creation of a new solution from scratch. This trend has limited the development of many other IoT based health care solutions that may constitute a higher added value if we were able to create new services by combining existing ones, benefiting from the interoperability that can become the driving force behind a new development impulse in highly heterogeneous contexts. Here we introduce a novel IoT design based health care monitoring system by using embedded controller with smart sensor network.

Objective

- 8 The primary objective of the system is to design an IoT as well as GSM platforms for remote health care monitoring system.
- 9 The secondary objective of the system is to design and implement the embedded programming structure for finding the human health parameters based on arduino environment with sensor network.

○ EXISTING SYSTEM

Blockchain technology (BT) has a wide range of built-in features, such as decentralization, transparency, data provenance, security, immutability, and has moved beyond the hype to practical applications in the healthcare industry. Telehealth turns out to be the most efficient and effective way of dispensing healthcare services even in remote areas. Though telehealth has the proven potential to improve the quality of healthcare, its implementation and adoption remain far from ideal. There is a need for a telehealth system that not only ensures the privacy and security of the users but also provides authentic services that enhance the trust to the highest level. Existing applications are leveraging BT and providing limited telehealth services. These applications are focused on a few services; hence do not cover all aspects of healthcare. To unveil the true potential of telehealth, this paper aims to develop an effective telehealth framework-BlockHeal which integrates all essential healthcare services under one platform and ensures a full-edged trusted environment. The methodology employed includes survey of existing telehealth system, identifying their weaknesses and critical reasons behind their lack of widespread adoption. The proposed framework addresses the limitations and includes all stakeholders of the healthcare system instigating a consolidated platform to ensure authentic, safe, and timely healthcare facilities. Additionally, as the proposed framework is based on BT, it ensures the provision of secure, fault-tolerant, transparent, and tamper-proof data. Moreover, it offers decentralized storage via hyperledger fabric and a collection of decentralized applications (DApps). Finally, the effectiveness of the BlockHeal framework is validated by demonstrating several use-cases.

Problem Statements

- Blockchain based Telehealth services need large bandwidth and network protocol.
- Need huge memory variables and constraints.
- High memory space requirement.
- Scalability issues, high energy consumption, potential inefficiencies, high implementation costs, and regulatory uncertainty, alongside concerns about privacy and security.

○ PROPOSED SYSTEM

The proposed system focused towards measuring the patient health parameters continuously by using wearable bio sensors such as IR based heart pulse rate detector, LDR based blood pressure sensing unit, NTC103 thermistor based body temperature detector and measured parameters are monitored through IOT device ESP8266. The embedded controller ATmega 328 can be used to process the entire monitoring system. The controller activates the buzzer for emergency local alert when the health parameters of the patient are in abnormal condition. The controller sends the emergency alert SMS to supervisor of the patient like doctor or hospital admin by using GSM800L Modem.

CHAPTER 2

LITERATURE SURVEY

- **Narmeen zakaria bawany , Tehreem Qamar, Hira Tariq , and Saifullah Adnan “Integrating Healthcare Services Using Blockchain-Based Telehealth Framework” Digital Object Identifier 10.1109/ACCESS.2022.3161944IEEE 2022**

Blockchain technology (BT) has a wide range of built-in features, such as decentralization, transparency, data provenance, security, immutability, and has moved beyond the hype to practical applications in the healthcare industry. Telehealth turns out to be the most efficient and effective way of dispensing healthcare services even in remote areas. Though telehealth has the proven potential to improve the quality of healthcare, its implementation and adoption remain far from ideal. There is a need for a telehealth system that not only ensures the privacy and security of the users but also provides authentic services that enhance the trust to the highest level. Existing applications are leveraging BT and providing limited telehealth services. These applications are focused on a few services; hence do not cover all aspects of healthcare. To unveil the true potential of telehealth, this paper aims to develop an effective telehealth framework- BlockHeal which integrates all essential healthcare services under one platform and ensures a full-edged trusted environment. The methodology employed includes survey of existing telehealth system, identifying their weaknesses and critical reasons behind their lack of widespread adoption. The proposed framework addresses the limitations and includes all stakeholders of the healthcare system instigating a consolidated platform to ensure authentic, safe, and timely healthcare facilities. Additionally, as the proposed framework is based on BT, it ensures the provision of secure, fault-tolerant, transparent, and tamper-proof data. Moreover, it offers decentralized storage via hyperledger fabric and a collection of centralized applications (DApps). Finally, the effectiveness of the BlockHeal framework is validated by demonstrating several use-cases.

- **Martin Jürgens*, Dennis Meis Bluetooth Mesh Networks for Indoor Localization International Conference on Mobile Data Management (MDM) IEEE 2019**

Indoor Localization Systems (ILS) are increasingly applied in various use cases, such as tracking the position and movement of staff or equipment within buildings. Creating and applying these systems requires decisions about measuring and transmitting signals, the algorithm transforming signals into position estimations as well as the representation displayed to the user. One approach gaining significance recently in this context are Mesh Networks. A Mesh Network enables single nodes to exchange

information with other nodes directly and dynamically, increases the error-tolerance, and facilitates the setup by reducing installation overhead. This paper studies the applicability of Bluetooth Mesh Networking technology for ILS by means of an experimental setup. The quality of localization is evaluated depending on different inference approaches as well as the scalability of the Bluetooth Mesh Technology. As initial results show, ILS based on Bluetooth Mesh Networks provide similar results regarding the accuracy of localization compared to other underlying technologies, such as Wi-Fi mesh networks, and can be set up with lower effort and costs. However, the underlying Bluetooth technology limits the scalability of the mesh network, especially in case of permanent localization of nodes.

- **Busra Ozlu¹, Bong Sup Shim Flexible and Wearable PEDOT-Paper Pressure Sensor for Detecting Human Voice IEEE -2021 international conference on Big Data and Smart Computing**

Flexible pressure sensors are gaining considerable attention as wearable devices to be potentially utilized in human machine interfaces. However, the preparation of pressure sensors through a simple, environmentally friendly, and cost-efficient fabrication process, remains as a challenge. In this study, we developed a flexible and wearable piezo resistive pressure sensor which consists of a polyethylene terephthalate (PET) as a flexible substrate for printing silver inter digitated electrodes and a poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) coated tissue paper as an active material. The PEDOT-Paper pressure sensor showed a sensitivity of 1.7/kPa in the range of 0 to 0.46 kPa, a response time of 240 ms and a recovery time of 306 ms. The potential application of this sensor for human voice detection was also demonstrated by mounting the sensor on the throat of a volunteer and pronouncing the words, _Hello_ and _Inha University_. The good repeatability and characteristic curves were achieved for each word, confirming the promising application of the PEDOT-Paper pressure sensor in the field of voice recognition.

- **Mahmood Khayatzadeh, Student Member, IEEEA 0.7-V 17.4-μW 3-Lead**

Wireless ECG SoCIEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS2013 IEEE

This paper presents a fully integrated sub-1 V 3-lead wireless ECG System-on-Chip (SoC) for wireless body sensor network applications. The SoC includes a two-channel ECG front-end with a driven-right-leg circuit, an 8-bit SAR ADC, a custom-designed 16-bit microcontroller, two banks of 16 kb SRAM, and a MICS band transceiver. The microcontroller and SRAM blocks are able to operate at sub-/near-threshold regime for the best energy consumption. The proposed SoC has been implemented in a standard 0.13- μ m CMOS process. Measurement results show the microcontroller consumes only 2.62 pJ per instruction at 0.35 V. Both microcontroller and memory blocks are functional down to 0.25 V. The entire SoC is capable of working at single 0.7-V supply. At the best case, it consumes 17.4 μ W in heart rate detection mode and 74.8 μ W in raw data acquisition mode under sampling rate of 500 Hz. This makes it one of the best ECG SoCs among state-of-the-art biomedical chips.

- **Prema Sundaram¹** Biomedical Engineering & P.S.N.A College of Engineering and Technology, India Patient monitoring system using android technology IJCSMC, Vol. 2, Issue. 5, May 2013

Telemedicine is a rapidly developing application of clinic medicine where medical information is transferred through the phone or internet or other networks for the purpose of consulting and performing remote medical procedures or examinations. Telemedicine can be applied to a greater extent in the field of cardiology where ECG serves as the major tool. This project elaborates the experience; a methodology adopted and highlights various design aspects to be considered for making telemedicine in patient monitoring system effective. In this method, the patient's vital signs like ECG, heart rate, breathing rate, temperature, SpO₂ are captured and the values are entered into the database. It is then uploaded into the web based server and sent to the doctor's phone using ANDROID technology. It also enables the doctors to instantly send back their feedback to the nurse station. The modern visionary of healthcare industry is to provide better healthcare to

people anytime and anywhere in the world in a more economic and patient friendly manner. Therefore for increasing the patient care efficiency, there arises a need to improve the patient monitoring devices and make them more mobile. The medical world today faces two basic problems when it comes to patient monitoring. Firstly, the needs of health care's provider's presence near the bedside of the patient and secondly, the patient is restricted to bed and wired to large machines. In order to achieve better quality patient care, the above cited problems have to be solved. As the bio instrumentation, computers and telecommunications technologies are advancing, it has become feasible to design more portal vital sign tele monitoring systems to acquire, record, display and to transmit the physiological signal from the human body to any location. Recent works in communication technologies have inspired the development of telemedicine to a large extent. Telemedicine benefits not only the customers who are able to receive health care more efficiently; it also benefits the doctors who can streamline their efforts to assist more patients.

- **Chin-Teng Lin, Fellow, IEEE, Kuan-Cheng Chang, Chun-Ling Lin An Intelligent Telecardiology System Using a Wearable and Wireless ECG to Detect Atrial Fibrillation IEEE TRANSACTIONS ON INFORMATION TECHNOLOGY IN BIOMEDICINE, VOL. 14, NO. 3, MAY 2010**

This study presents a novel wireless, ambulatory, real time, and auto alarm intelligent telecardiology system to improve healthcare for cardiovascular disease, which is one of the most prevalent and costly health problems in the world. This system consists of a lightweight and power-saving wireless ECG device equipped with a built-in automatic warning expert system. This device is connected to a mobile and ubiquitous real-time display platform. The acquired ECG signals are instantaneously transmitted to mobile devices, such as net books or mobile phones through Bluetooth, and then, processed by the expert system. An alert signal is sent to the remote database server, which can be accessed by an Internet browser, once an abnormal ECG is detected. The current version of the expert system can identify five types of abnormal cardiac rhythms in real-time, including sinus tachycardia, sinus bradycardia, wide QRS complex, atrial fibrillation (AF), and cardiac asystole, which is very important for both the subjects who

are being monitored and the healthcare personnel tracking cardiac-rhythm disorders. The proposed system also activates an emergency medical alarm system when problems occur. Clinical testing reveals that the proposed system is approximately 94% accurate, with high sensitivity, specificity, and positive prediction rates for ten normal subjects and 20 AF patients. We believe that in the future a business-card-like ECG device, accompanied with a mobile phone, can make universal cardiac protection service possible.

- **Joan Gómez-Clapers, Ramon Casanella Wireless ECG and Heart Rate Monitoring Using Dual Ground Dry Electrodes Instrumentation for the ICT Era Sept. 8-10, 2010, Kosice, Slovakia Castelldefels (Barcelona) – SPAIN**

In this work, a novel wireless easy-to-use measuring system is presented to acquire the Lead I ECG signal and the heart rate. The system presents a novel application of the dual ground configuration to dry electrodes, to reduce their level of power line interference. With this, a good quality Lead I EGC signal can be obtained simply by placing the right and left hands on the dual electrodes. Heart rate is obtained with a novel algorithm based on the continuous wavelet transform (CWT), especially designed to avoid the electromyographic noise that can be present when acquiring signals in the hands. The algorithm presented has been tested in twelve subjects of different age and physical condition, obtaining a 99.7% of sensitivity and a 100 % of positive predictivity.

CHAPTER 3

PROPOSED SYSTEM FUNCTION

3.1 SYSTEM OPERATION

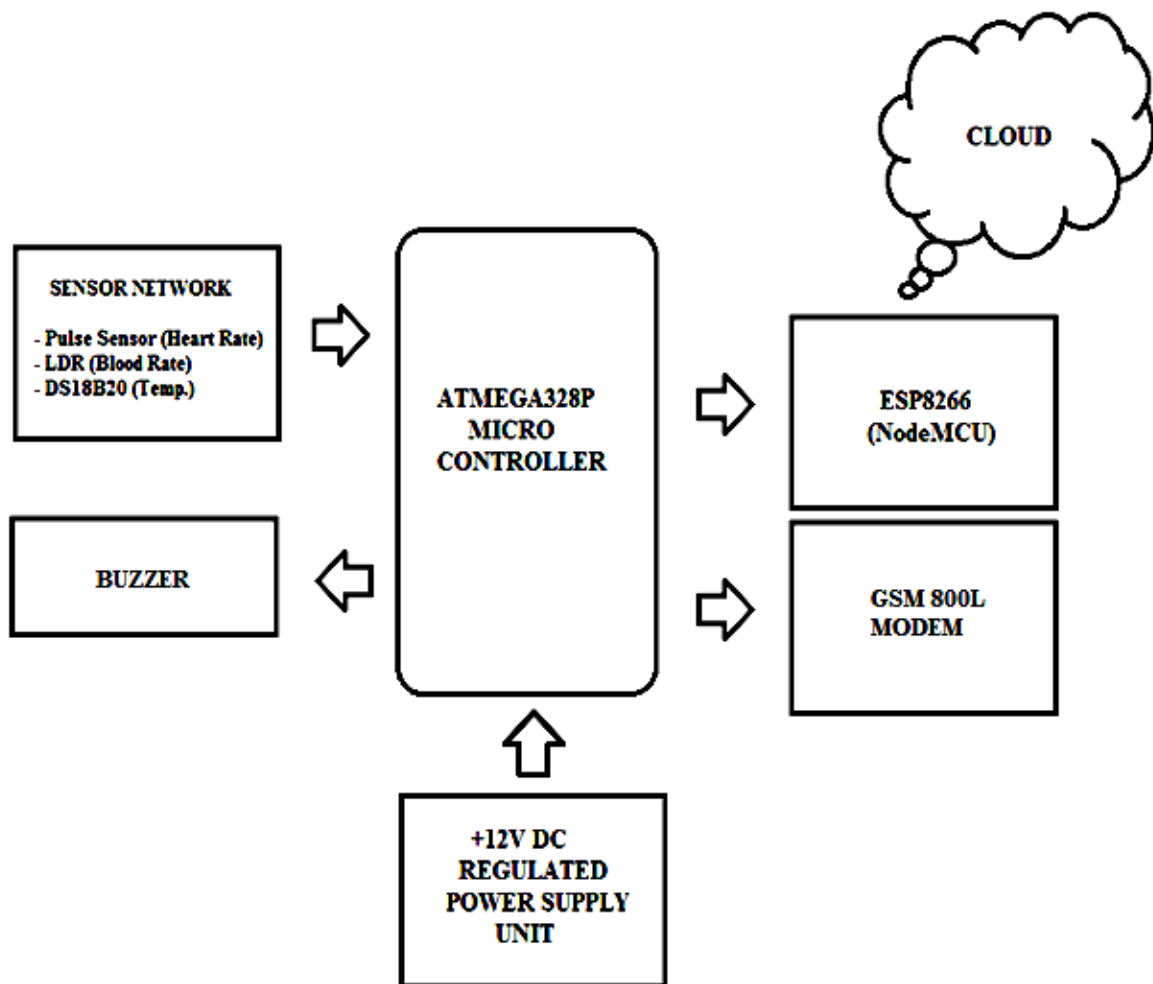


Fig.3.1 Block diagram of the system

In this system consists of ATMEGA328P microcontroller (arduino uno R3), pulse sensor, LDR (BP), NTC-103 temperature sensor, Buzzer, GSM800L MODEM, ESP8266 (NodeMCU-WiFi module) and +12VDC / +9VDC regulated power supply units. The

pulse sensor is used to detect the heart rate pulse, NTC-103 (Thermistor) is used to finding the temperature level of the human body and it generates 10mV/ 1 degree Celsius. The LDR based blood rate module is used to detect the blood pressure rate of human body. All the sensors outputs are in the form of electrical signal according to their input physical signal.

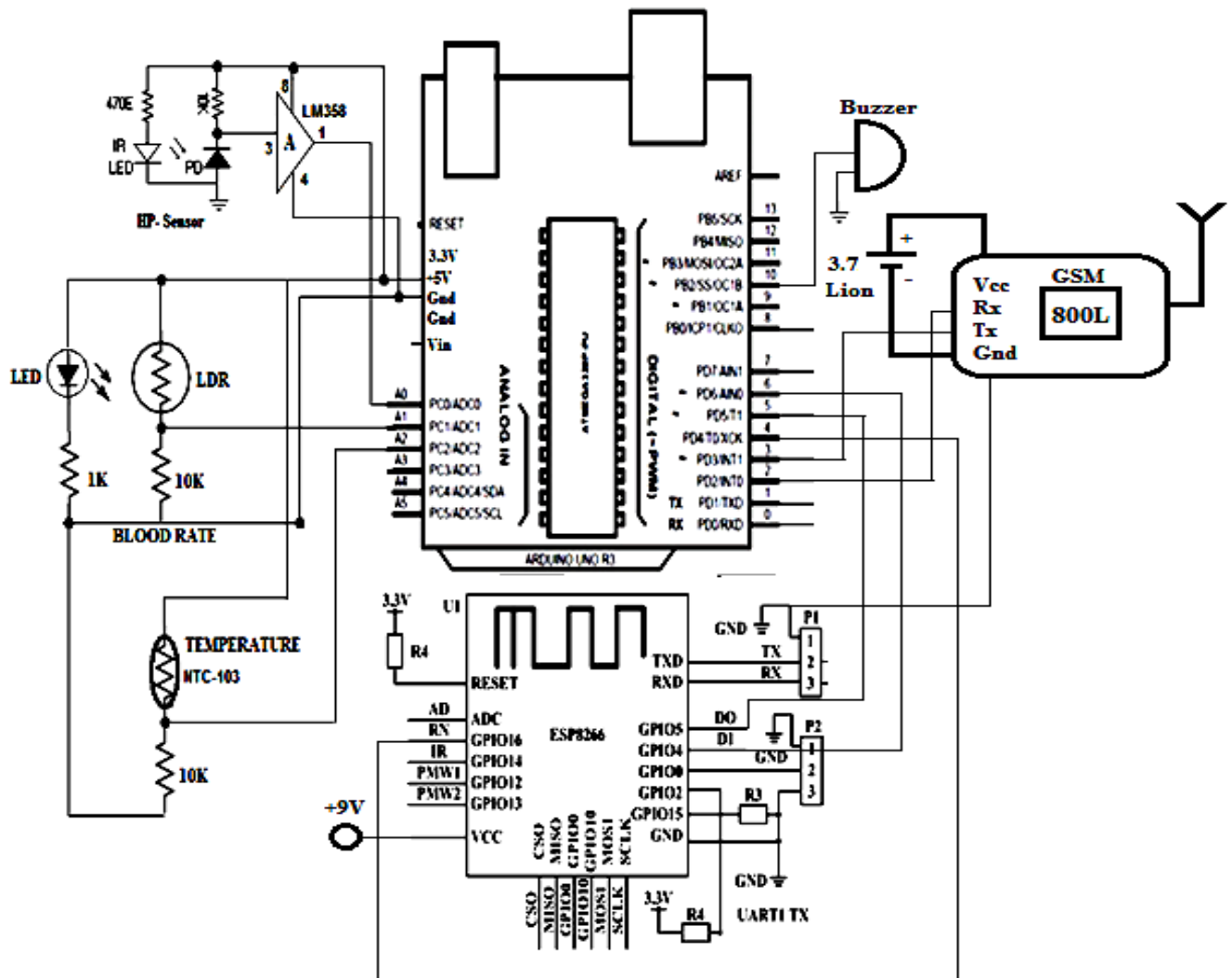


Fig.3.2 circuit diagram of the system

All the sensor outputs (electrical) are applied to the ADC (analog to digital converter) input of microcontroller. The controller executes the predesigned program to calculate the sensor values and it send to ESP8266. The ESP8266 is a WiFi module which is used to connect the controller data into CLOUD network (INTERNET). The data about the health care system can be monitored anywhere around the world through internet. The controller activates the buzzer when the health parameters exceed predesigned reference value called abnormal health condition and also send the emergency alert via SMS to corresponding

admin using through GSM network.

The +12VDC regulated power supply generates the power for controller operation. +9V power supply unit is used to supplies the electrical power for IOT device (ESP8266 WiFi module) and 3.7V lithium ion battery is used for GSM MODEM.

3.2 POWER SUPPLY UNIT

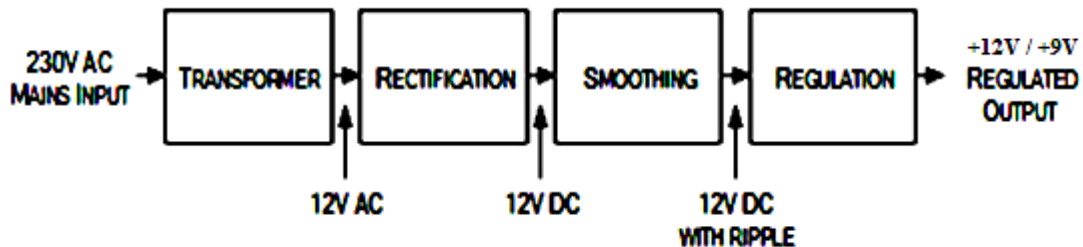


Fig: 3.3 Block diagram of power supply unit

AC power is easily in bulk from through different methods, but generally for many power control circuits and other industrial application DC power is very much required.

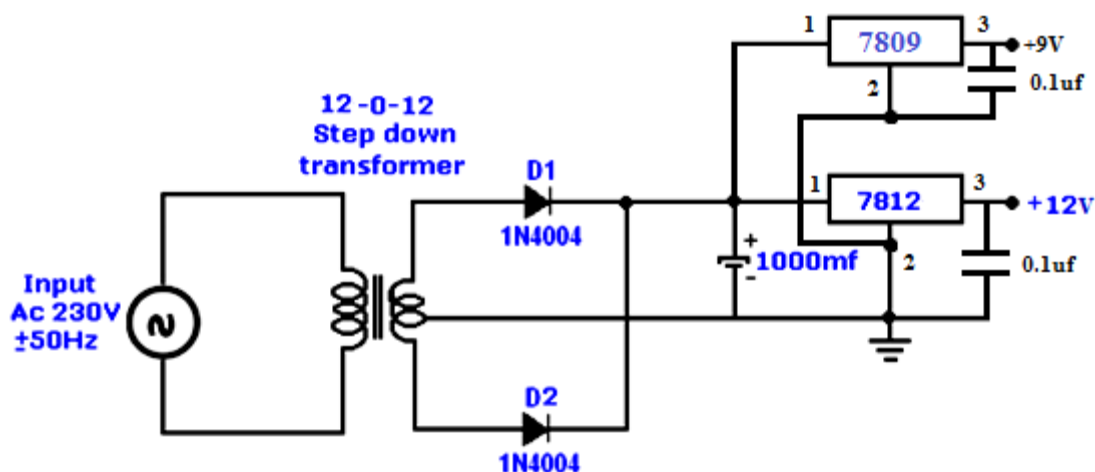


Fig: 3.4 Circuit diagram of power supply unit

Hence AC power necessarily has to be converted into DC power by means of electronic rectifier, which is simpler, cheaper, and highly efficient compared to rotary converters or DC generators. The “**rectifier**” is a circuit, which converts AC Voltage and currents into pulsating DC voltages and currents. It consists of DC components and the unwanted ac ripple or harmonic components, which can be removed by using filter circuit. Thus the output obtained will be steady DC voltage and magnitude of DC voltage can be varied by varying the magnitude of AC Voltage.

CHAPTER 4 HARDWARE DETAILS

4.1 ARDUINO-UNO-R3



Fig:3.1 Arduino uno R3 microcontroller

The **Arduino Uno R3** is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter. Revision 2 of the Uno board has a resistor pulling the 8U2 HWB line to ground, making it easier to put into DFU mode.

Revision 3 of the board has the following new features:

- 1.0 pinout: added SDA and SCL pins that are near to the AREF pin and two other new pins placed near to the RESET pin, the IOREF that allow the shields to adapt to the voltage provided from the board. In future, shields will be compatible both

with the board that use the AVR, which operate with 5V and with the Arduino Due that operate with 3.3V. The second one is a not connected pin that is reserved for future purposes.

- Stronger RESET circuit.
- Atmega 16U2 replace the 8U2.

"Uno" means one in Italian and is named to mark the upcoming release of Arduino 1.0. The Uno and version 1.0 will be the reference versions of Arduino, moving forward. The Uno is the latest in a series of USB Arduino boards, and the reference model for the Arduino platform; for a comparison with previous versions, see the index of Arduino boards.

TABLE: 1 Summary

Microcontroller	ATmega328
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limits)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	40 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB (ATmega328) of which 0.5 KB used by bootloader
SRAM	2 KB (ATmega328)
EEPROM	1 KB (ATmega328)
Clock Speed	16 MHz

The Arduino reference design can use an Atmega8, 168, or 328, Current models use an ATmega328, but an Atmega8 is shown in the schematic for reference. The pin configuration is identical on all three processors.

4.1.1 Power

The Arduino Uno can be powered via the USB connection or with an external power supply. The power source is selected automatically. External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. The adapter can be connected by plugging a 2.1mm center-positive plug into the board's power jack. Leads from a battery can be inserted in the Gnd and Vin pin headers of the POWER connector. The board can operate on an external supply of 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may be unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

4.1.2 The power pins are as follows:

- **VIN.** The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.
- **5V.** The regulated power supply used to power the microcontroller and other components on the board. This can come either from VIN via an on-board regulator, or be supplied by USB or another regulated 5V supply.
- **3V3.** A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- **GND.** Ground pins.

4.1.3 Memory

The ATmega328 has 32 KB (with 0.5 KB used for the bootloader). It also has 2 KB of SRAM and 1 KB of EEPROM (which can be read and written with the EEPROM library).

4.1.4 Input and Output

Each of the 14 digital pins on the Uno can be used as an input or output, using `pinMode()`, `digitalWrite()`, and `digitalRead()` functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40 mA and has an internal pull-up resistor (disconnected by default) of 20-50 kOhms. In addition, some pins have specialized functions:

1. **Serial: 0 (RX) and 1 (TX).** Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip.
2. **External Interrupts: 2 and 3.** These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value. See the `attachInterrupt()` function for details.
3. **PWM: 3, 5, 6, 9, 10, and 11.** Provide 8-bit PWM output with the `analogWrite()` function.
4. **SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK).** These pins support SPI communication using the SPI library.
5. **LED: 13.** There is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

The Uno has 6 analog inputs, labeled A0 through A5, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from ground to 5 volts, though it is possible to change the upper end of their range using the AREF pin and the `analogReference()` function. Additionally, some pins have specialized functionality:

- **TWI: A4 or SDA pin and A5 or SCL pin.** Support TWI communication using the Wire library.

There are a couple of other pins on the board:

- **AREF.** Reference voltage for the analog inputs. Used with `analogReference()`.
- **Reset.** Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

See also the mapping between Arduino pins and ATmega328 ports. The mapping for the ATmega8, 168, and 328 is identical.

4.1.5 Communication

The Arduino Uno has a number of facilities for communicating with a computer, another Arduino, or other microcontrollers. The ATmega328 provides UART TTL (5V) serial communication, which is available on digital pins 0 (RX) and 1 (TX). An ATmega16U2 on the board channels this serial communication over USB and appears as a virtual com port to software on the computer. The '16U2 firmware uses the standard USB COM drivers, and no external driver is needed. However, on Windows, a .inf file is required. The Arduino software includes a serial monitor which allows simple textual data to be sent to and from the Arduino board. The RX and TX LEDs on the board will flash when data is being transmitted via the USB-to-serial chip and USB connection to the computer (but not for serial communication on pins 0 and 1). A SoftwareSerial library allows for serial communication on any of the Uno's digital pins. The ATmega328 also supports I2C (TWI) and SPI communication. The Arduino software includes a Wire library to simplify use of the I2C bus; see the documentation for details. For SPI communication, use the SPI library.

4.1.6 Programming

The Arduino Uno can be programmed with the Arduino software (download). Select "Arduino Uno" from the **Tools > Board** menu (according to the microcontroller on your board). For details, see the reference and tutorials. The ATmega328 on the Arduino Uno comes preburned with a bootloader that allows you to upload new code to it without the use of an external hardware programmer. It communicates using the original STK500 protocol (reference, C header files). You can also bypass the bootloader and program the microcontroller through the ICSP (In-Circuit Serial Programming) header; see these instructions for details.

The ATmega16U2 (or 8U2 in the rev1 and rev2 boards) firmware source code is available . The ATmega16U2/8U2 is loaded with a DFU bootloader, which can be activated by:

- On Rev1 boards: connecting the solder jumper on the back of the board (near the map of Italy) and then resetting the 8U2.
- On Rev2 or later boards: there is a resistor that pulling the 8U2/16U2 HWB line to ground, making it easier to put into DFU mode.

Automatic (Software) Reset

Rather than requiring a physical press of the reset button before an upload, the Arduino Uno is designed in a way that allows it to be reset by software running on a connected computer. One of the hardware flow control lines (DTR) of the ATmega8U2/16U2 is connected to the reset line of the ATmega328 via a 100 nanofarad capacitor. When this line is asserted (taken low), the reset line drops long enough to reset the chip. The Arduino software uses this capability to allow you to upload code by simply pressing the upload button in the Arduino environment. This means that the bootloader can have a shorter timeout, as the lowering of DTR can be well-coordinated with the start of the upload.

This setup has other implications. When the Uno is connected to either a computer running Mac OS X or Linux, it resets each time a connection is made to it from software (via USB). For the following half-second or so, the bootloader is running on the Uno. While it is programmed to ignore malformed data (i.e. anything besides an upload of new code), it will intercept the first few bytes of data sent to the board after a connection is opened. If a sketch running on the board receives one-time configuration or other data when it first starts, make sure that the software with which it communicates waits a second after opening the connection and before sending this data. The Uno contains a trace that can be cut to disable the auto-reset. The pads on either side of the trace can be soldered together to re-enable it. It's labeled "RESET-EN". You may also be able to disable the auto-reset by connecting a 110 ohm resistor from 5V to the reset line; see this forum thread for details.

4.1.7 USB Over current Protection

The Arduino Uno has a resettable polyfuse that protects your computer's USB ports from shorts and overcurrent. Although most computers provide their own internal

protection, the fuse provides an extra layer of protection. If more than 500 mA is applied to the USB port, the fuse will automatically break the connection until the short or overload is removed.

4.1.8 Physical Characteristics

The maximum length and width of the Uno PCB are 2.7 and 2.1 inches respectively, with the USB connector and power jack extending beyond the former dimension. Four screw holes allow the board to be attached to a surface or case. Note that the distance between digital pins 7 and 8 is 160 mil (0.16"), not an even multiple of the 100 mil spacing of the other pins.

4.2 SENSORS

4.2.1 PULSE SENSOR



Fig: 4.2 schematic diagram of IR LED

Infrared (IR) light is electromagnetic radiation with a wavelength between 0.7 and 300 micrometres, which equates to a frequency range between approximately 1 and 430 THz. IR wavelengths are longer than that of visible light, but shorter than that of terahertz radiation microwaves. Bright sunlight provides an irradiance of just over 1 kilowatt per square meter at sea level. Of this energy, 527 watts is infrared radiation, 445 watts is visible light, and 32 watts is ultraviolet radiation.

IR data transmission is also employed in short-range communication among computer peripherals and personal digital assistants. These devices usually conform to standards published by IrDA, the Infrared Data Association. Remote controls and IrDA devices use infrared light-emitting diodes (LEDs) to emit infrared radiation which is focused by a plastic lens into a narrow beam. The beam is modulated, i.e. switched on and

off, to encode the data. The receiver uses a silicon photodiode to convert the infrared radiation to an electric current. It responds only to the rapidly pulsing signal created by the transmitter, and filters out slowly changing infrared radiation from ambient light. Infrared communications are useful for indoor use in areas of high population density. IR does not penetrate walls and so does not interfere with other devices in adjoining rooms. Infrared is the most common way for remote controls to command appliances.

4.2.2 IR RECEIVING SENSOR



Fig: 4.3 schematic diagram of IR Photodiode

A **photodiode** is a type of photo detector capable of converting light into either current or voltage, depending upon the mode of operation. Photodiodes are similar to regular semiconductor diodes except that they may be either exposed (to detect vacuum UV or X-rays) or packaged with a window or optical fiber connection to allow light to reach the sensitive part of the device. Many diodes designed for use specifically as a photodiode will also use a PIN junction rather than the typical PN junction.

Principle of operation

A photodiode is a PN junction or PIN structure. When a photon of sufficient energy strikes the diode, it excites an electron, thereby creating a *free electron* and a (positively charged electron) *hole*. If the absorption occurs in the junction's depletion region, or one diffusion length away from it, these carriers are swept from the junction by the built-in field of the depletion region. Thus holes move toward the anode, and electrons toward the cathode, and a photocurrent is produced.

Photovoltaic mode

When used in zero bias or *photovoltaic mode*, the flow of photocurrent out of the device is restricted and a voltage builds up. The diode becomes forward biased and "dark current" begins to flow across the junction in the direction opposite to the photocurrent. This mode is responsible for the photovoltaic effect, which is the basis for solar cells – in fact, a traditional solar cell is just a large area photodiode.

Photoconductive mode

In this mode the diode is often reverse biased, dramatically reducing the response time at the expense of increased noise. This increases the width of the depletion layer, which decreases the junction's capacitance resulting in faster response times. The reverse bias induces only a small amount of current (known as saturation or back current) along its direction while the photocurrent remains virtually the same. For a given spectral distribution, the photocurrent is linearly proportional to the luminance (and to the irradiance).^[2]

Although this mode is faster, the photoconductive mode tends to exhibit more electronic noise. The leakage current of a good PIN diode is so low ($< 1\text{nA}$) that the Johnson–Nyquist noise of the load resistance in a typical circuit often dominates.

Advantages:

- Excellent linearity of output current as a function of incident light
- Spectral response from 190 nm to 1100 nm (silicon), longer wavelengths with other semiconductor materials
- Low noise
- Ruggedized to mechanical stress
- Low cost
- Compact and light weight
- Long lifetime
- High quantum efficiency, typically 80%
- No high voltage required

Working of the Pulse Sensor

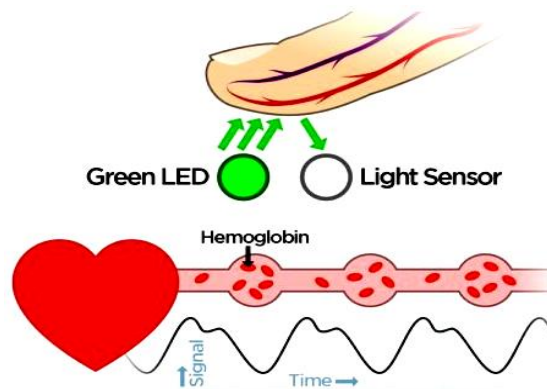


Fig.4.4: working principle of the pulse sensor

The Pulse Sensor works on the principle of Photoplethysmography (PPG), which is a non-invasive method for measuring changes in blood volume under the skin. The sensor essentially consists of two main components: a light-emitting diode (LED) that shines light into the skin and a photodetector that measures the amount of light that is reflected back. Here's a detailed explanation of its working:

- **Light Emission:** A green LED emits light into the skin.
- **Reflection & Detection:** The light interacts with blood and is partially reflected back, captured by a photodetector.
- **Heart Rate:** Changes in reflected light create a waveform that correlates with heartbeats.
- **Oxygen Level:** The amount of reflected light also indicates blood oxygen levels, as oxygenated blood absorbs more green light.
- **Signal Filtering:** A Low Pass Filter cleans up the noisy, raw signal from the photodetector.
- **Amplification:** An operational amplifier boosts the filtered signal for better

accuracy.

- **Data Reading:** Finally, an Arduino reads the amplified signal and software algorithms translate it into heart rate and potentially blood oxygen levels.

4.2.3 LDR: LIGHT DEPENDENT RESISTOR

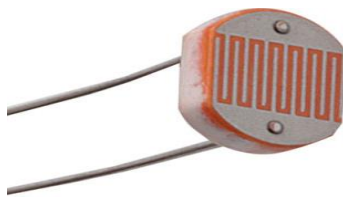


Fig.4.5 schematic of LDR and symbol

LDRs or Light Dependent Resistors are very useful especially in light/dark sensor circuits. Normally the resistance of an LDR is very high, sometimes as high as 1000000 ohms, but when they are illuminated with light resistance drops dramatically. Electronic sensors are the devices that alter their electrical characteristics, in the presence of visible or invisible light. The best-known devices of this type are the light dependent resistor (LDR), the photo diode and the phototransistors.

Light dependent resistor as the name suggests depends on light for the variation of resistance.

- LDR are made by depositing a film of cadmium sulphide or cadmium selenide on a substrate of ceramic containing no or very few free electrons when not illuminated. The longer the strip the more the value of resistance.
- When light falls on the strip, the resistance decreases. In the absence of light the resistance can be in the order of 10K ohm to 15K ohm and is called the dark resistance.

Depending on the exposure of light the resistance can fall down to value of 500 ohms. The power ratings are usually smaller and are in the range 50mw to .5w. Though very sensitive to light, the switching time is very high and hence cannot be used for high frequency applications. They are used in chopper amplifiers. Light dependent resistors are available

as discs 0.5cm to 2.5cm. The resistance rises to several Mega ohms under dark conditions.

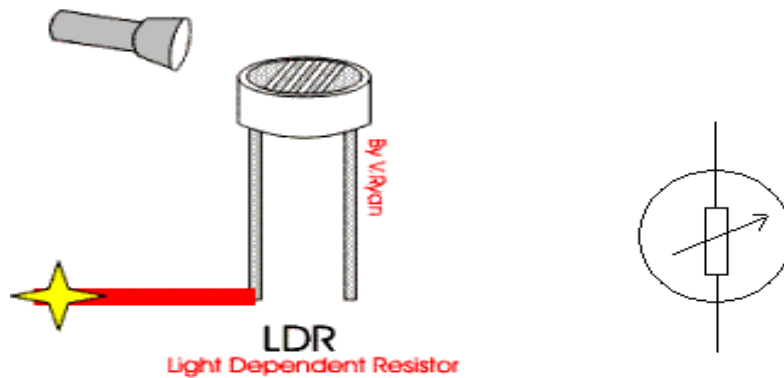


Fig.4.6 LDR Operation

The below figure shows that when the torch is turned on, the resistance of the LDR falls, allowing current to pass through it is shown in figure. The basic construction and symbol for LDR are shown in above figures respectively. The device consists of a pair of metal film contacts separated by a snakelike track of cadmium sulphide film, designed to provide the maximum possible contact area with the two metal films. The structure is housed in a clear plastic or resin case, to provide free access to external light. Practical LDRs are available in variety of sizes and packages styles, the most popular size having a face diameter of roughly 10mm. practical LDR is shown in below figure.

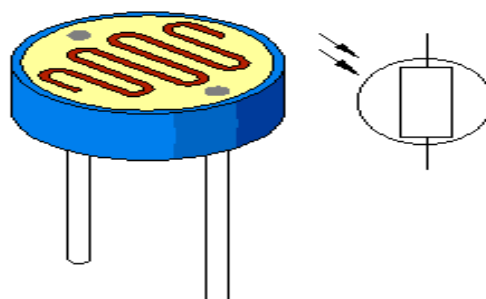


Fig.4.7 Construction of LDR

Recovery rate:

When an LDR is brought from a certain illuminating level into total darkness, the resistance does not increase immediately to the dark value. The recovery rate is specified in k ohm/second and for current LDR types it is more than 200k ohm/second. The recovery rate is much greater in the reverse direction, e.g. going from darkness to

illumination level of 300 lux, it takes less than 10ms to reach a resistance which corresponds with a light level of 400 lux. A LDR may be connected either way round and no special precautions are required when soldering.

- Darkness: Maximum resistance, about 1Mohm.
- Very bright light: Minimum resistance, about 100 ohm.

The LDR is a variable resistor whose resistance decreases with the increase in light intensity. Two cadmium sulphide (cds) photoconductive cells with spectral response similar to that of the human eye. The cell resistance falls with increasing light intensity. some of its features:

- High reliability.
- Light weight.
- Wide spectral response.
- Wide ambient temperature range.

Applications

Photoresistors come in many different types. Inexpensive cadmium sulfide cells can be found in many consumer items such as camera light meters, street lights, clock radios, alarms, and outdoor clocks.

They are also used in some dynamic compressors together with a small incandescent lamp or light emitting diode to control gain reduction. Lead sulfide (PbS) and indium antimonide (InSb) LDRs (light dependent resistor) are used for the mid infrared spectral region. Ge:Cu photoconductors are among the best far-infrared detectors available, and are used for infrared astronomy and infrared spectroscopy.

4.2.4 THERMISTOR – NTC 103

A thermistor is a type of resistor whose resistance varies significantly with temperature, more so than in standard resistors. The word is a portmanteau of thermal and resistor. Thermistors are widely used as inrush current limiters, temperature sensors, self-

resetting over current protectors, and self-regulating heating elements.

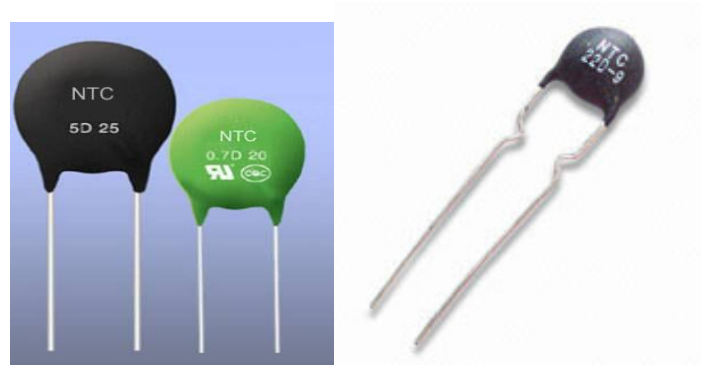


Fig..4.8 schematic diagram of thermistor

BASIC OPERATION

Assuming, as a first-order approximation, that the relationship between resistance and temperature is linear, then:

$$\Delta R = k\Delta T$$

where

ΔR = change in resistance

ΔT = change in temperature

k = first-order temperature coefficient of resistance

Thermistors can be classified into two types, depending on the sign of k . If k is positive, the resistance increases with increasing temperature, and the device is called a positive temperature coefficient (PTC) thermistor, or posistor. If k is negative, the resistance decreases with increasing temperature, and the device is called a negative temperature coefficient (NTC) thermistor. Resistors that are not thermistors are designed to have a k as close to zero as possible, so that their resistance remains nearly constant over a wide temperature range. Instead of the temperature coefficient k , sometimes the temperature coefficient of resistance α_T (alpha sub T) is used. It is defined as

$$\alpha_T = \frac{1}{R(T)} \frac{dR}{dT}.$$

This α_T coefficient should not be confused with the a parameter below.

WORKING OF THERMISTOR:

A thermistor is simply an electrical resistor whose resistance changes rapidly with temperature. Usually thermistors are made from small beads of complex materials and

although they are not as reproducible as platinum resistance thermometers, they are much more sensitive. Commonly the resistance falls exponentially with increasing temperature and such devices are said to have a negative temperature coefficient (NTC). The strong change in temperature is useful for establishing sensitive temperature control of a system.

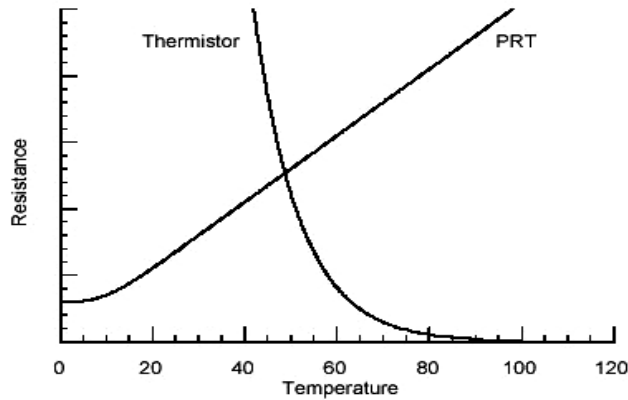


Fig.4.9 Comparison graph Resistance Vs Temperature

As a consequence of the high sensitivity, the range of use of an individual thermistor is comparatively narrow, and a variety of thermistors are needed to suit different ranges and applications. They are popular in digital thermometers for everyday, medical and industrial use. The following graph compares the characteristic of a low-temperature thermistor with that of a PRT.

4.3 GSM800L

A GSM modem is a wireless modem that works with a GSM wireless network. A wireless modem behaves like a dial-up modem. The main difference between them is that a dial-up modem sends and receives data through a fixed telephone line while a wireless modem sends and receives data through radio waves. A GSM modem can be an external device or a PC Card / PCMCIA Card. Typically, an external GSM modem is connected to a computer through a serial cable or a USB cable. A GSM modem in the form of a PC Card / PCMCIA Card is designed for use with a laptop computer. It should be inserted into one of the PC Card / PCMCIA Card slots of a laptop computer. Like a GSM mobile phone, a GSM modem requires a SIM card from a wireless carrier in order to operate.

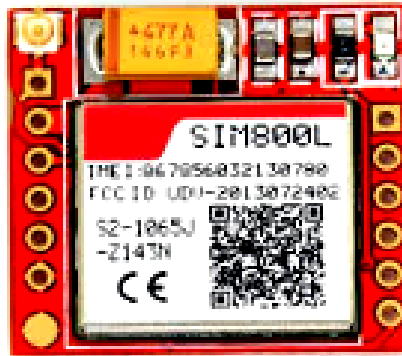


Fig: 4.10 Schematic diagram of GSM MODEM

As mentioned in earlier sections of this SMS tutorial, computers use AT commands to control modems. Both GSM modems and dial-up modems support a common set of standard AT commands. You can use a GSM modem just like a dial-up modem. In addition to the standard AT commands, GSM modems support an extended set of AT commands. These extended AT commands are defined in the GSM standards. With the extended AT commands, you can do things like:

- Reading, writing and deleting SMS messages.
- Sending SMS messages.
- Monitoring the signal strength.
- Monitoring the charging status and charge level of the battery.
- Reading, writing and searching phone book entries.
- The number of SMS messages that can be processed by a GSM modem per minute is very low -- only about six to ten SMS messages per minute.

AT-Command set

The following section describes the AT-Command set. The commands can be tried out by connecting a GSM modem to one of the PC's COM ports. Type in the test-command, adding CR + LF (Carriage return + Line feed = `\r\n`) before executing. Also see

chapter 3.1 for further details. Table 2-1 gives an overview of the implemented AT-Commands in this application. The use of the commands is described in the later sections.

- The following formats are used in Table 2-2 through 2-9:
- Character string in quotation marks is the actual text sent to modem.
- Optional commands and response parameters are enclosed in brackets.

Status (AT) the “AT” command is a status request used for testing if a compatible modem is connected and that the serial interface is working properly.

Echo off (ATE0)

The “ATE0” command is used to config the communication. By default, GSM modems are set to echo any received command back with an acknowledgement. An example of this is shown below.

- AT\r\n //Command sent to modem
- AT\r\nOK\r\n //Response from modem with echo enabled
- After sending “AT”, the modem replies with “AT\r\nOK\r\n”. With echo off, “ATE0”, the modem would have answered “\r\nOK\r\n” when executing “AT”.
- The echo off command will reduce traffic on the serial line. The “ATE1” command will enable echo again.

4.4 NODEMCU



Fig.4.11 Schematic of NODEMCU Module

ESP-12E WiFi module is developed by Ai-thinker Team. core processor NODEMCU in smaller sizes of the module encapsulates Tensilica L106 integrates industry-leading ultra low power 32-bit MCU micro, with the 16-bit short mode, Clock speed support 80 MHz, 160 MHz, supports the RTOS, integrated Wi-Fi MAC/BB/RF/PA/LNA, on-board antenna. The module supports standard IEEE802.11 b/g/n agreement, complete TCP/IP protocol stack. Users can use the add modules to an existing device networking, or building a separate network controller. NODEMCU is high integration wireless SOCs, designed for space and power constrained mobile platform designers. It provides unsurpassed ability to embed Wi-Fi capabilities within other systems, or to function as a standalone application, with the lowest cost, and minimal space requirement.

NODEMCUEX offers a complete and self-contained Wi-Fi networking solution; it can be used to host the application or to offload Wi-Fi networking functions from another application processor. When NODEMCUEX hosts the application, it boots up directly from an external flash. In has integrated cache to improve the performance of the system in such applications. Alternately, serving as a Wi-Fi adapter, wireless internet access can be added to any micro controller based design with simple connectivity (SPI/SDIO or I2C/UART interface).

Features

- 802.11 b/g/n
- Integrated low power 32-bit MCU, 10-bit ADC, TCP/IP protocol stack
- Integrated PLL, regulators, and power management units
- Supports antenna diversity
- Wi-Fi 2.4 GHz, support WPA/WPA2
- Support STA/AP/STA+AP operation modes
- Support Smart Link Function for both Android and iOS devices
- SDIO 2.0, (H) SPI, UART, I2C, I2S, IRDA, PWM, GPIO
- STBC, 1x1 MIMO, 2x1 MIMO

- A-MPDU & A-MSDU aggregation and 0.4s guard interval
- Deep sleep power <10uA, Power down leakage current < 5uA
- Wake up and transmit packets in < 2ms
- Standby power consumption of < 1.0mW (DTIM3)
- +20dBm output power in 802.11b mode
- Operating temperature range -40C ~ 125C

4.5 TRANSFORMER



Fig: 4.12 Schematic of transformer

A **transformer** is a static device that transfers electrical energy from one circuit to another through inductively coupled conductors—the transformer's coils. A varying current in the first or *primary* winding creates a varying magnetic flux in the transformer's core and thus a varying magnetic field through the *secondary* winding. This varying magnetic field induces a varying electromotive force (EMF) or "voltage" in the secondary winding. This effect is called mutual induction.

If a load is connected to the secondary, an electric current will flow in the secondary winding and electrical energy will be transferred from the primary circuit through the transformer to the load. In an ideal transformer, the induced voltage in the secondary winding (V_s) is in proportion to the primary voltage (V_p), and is given by the ratio of the number of turns in the secondary (N_s) to the number of turns in the primary (N_p) as follows:

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

BASIC PRINCIPLES

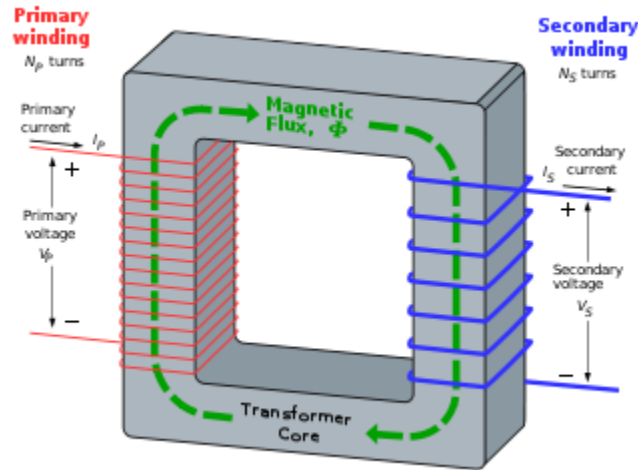


Fig: 4.13 Transformer principles

The transformer is based on two principles: first, that an electric current can produce a magnetic field (electromagnetism), and, second that a changing magnetic field within a coil of wire induces a voltage across the ends of the coil (electromagnetic induction). Changing the current in the primary coil changes the magnetic flux that is developed. The changing magnetic flux induces a voltage in the secondary coil.

Induction law

The voltage induced across the secondary coil may be calculated from Faraday's law of induction, which states that:

$$V_s = N_s \frac{d\Phi}{dt},$$

where V_s is the instantaneous voltage, N_s is the number of turns in the secondary coil and Φ is the magnetic flux through one turn of the coil. If the turns of the coil are oriented perpendicular to the magnetic field lines, the flux is the product of the magnetic flux density B and the area A through which it cuts. The area is constant, being equal to the

cross-sectional area of the transformer core, whereas the magnetic field varies with time according to the excitation of the primary. Since the same magnetic flux passes through both the primary and secondary coils in an ideal transformer, the instantaneous voltage across the primary winding equals. Taking the ratio of the two equations for V_s and V_p gives the basic equation for stepping up or stepping down the voltage

$$\frac{V_s}{V_p} = \frac{N_s}{N_p}$$

N_p/N_s is known as the *turns ratio*, and is the primary functional characteristic of any transformer. In the case of step-up transformers, this may sometimes be stated as the reciprocal, N_s/N_p . *Turns ratio* is commonly expressed as an irreducible fraction or ratio: for example, a transformer with primary and secondary windings of, respectively, 100 and 150 turns is said to have a turns ratio of 2:3 rather than 0.667 or 100:150.

4.6 Li ION BATTERY



Fig: 4.14 Schematic of Li ion battery (3.7V/4.4MH)

A **lithium-ion** or **Li-ion battery** is a type of rechargeable battery that uses the reversible intercalation of Li^+ ions into electronically conducting solids to store energy. In comparison with other commercial rechargeable batteries, Li-ion batteries are characterized by higher specific energy, higher energy density, higher energy efficiency, a longer cycle life, and a longer calendar life. Also noteworthy is a dramatic improvement in lithium-ion battery properties after their market introduction in 1991: over the following 30 years, their volumetric energy density increased threefold while their cost dropped

tenfold.^[8] In late 2024 global demand passed 1 Terawatt-hour per year,^[9] while production capacity was more than twice that. Generally, lithium-ion batteries are lighter and can be charged more rapidly than lead-acid batteries. And lithium-ion batteries are more environmentally friendly since they don't contain any substance with a high environmental load. Besides, lead-acid batteries are less costly. I've been using them for decades. The typical estimated life of a Lithium-Ion battery is about two to three years or 300 to 500 charge cycles, whichever occurs first. One charge cycle is a period of use from fully charged, to fully discharged, and fully recharged again.

- Capacity: 4400 mAh Capacity
- Cell with Voltage: 3.7 Volts
- Cell, Size: Diameter- 18mm x Length- 65mm. (Standard- 18650)
- Application: Small DRONES, GPS, Mp4 player, Power bank,
- This li-ion Cell is very convenient to install in your project where 3.7 Volt with high capacity is needed.
- The standard 18650 4400 mAh 3.7V rechargeable li-ion Cell is a source of power for various portable devices including flashlights.

4.7 BUZZER



Fig. 4.15 schematic of cam buzzer

Piezoelectric sensors have proven to be versatile tools for the measurement of various processes. They are used for quality assurance, process control and for research and development in many different industries. Although the piezoelectric effect was discovered by Curie in 1880, it was only in the 1950s that the piezoelectric effect started to

be used for industrial sensing applications. Since then, this measuring principle has been increasingly used and can be regarded as a mature technology with an outstanding inherent reliability. It has been successfully used in various applications, such as in medical, aerospace, nuclear instrumentation, and as a pressure sensor in the touch pads of mobile phones. In the automotive industry, piezoelectric elements are used to monitor combustion when developing internal combustion engines. The sensors are either directly mounted into additional holes into the cylinder head or the spark/glow plug is equipped with a built in miniature piezoelectric sensor.

The rise of piezoelectric technology is directly related to a set of inherent advantages. The high modulus of elasticity of many piezoelectric materials is comparable to that of many metals and goes up to $10 \times 10^6 \text{ N/m}^2$. Even though piezoelectric sensors are electromechanical systems that react to compression, the sensing elements show almost zero deflection.

4.8 IC7809 – POSITIVE VOLTAGE REGULATOR

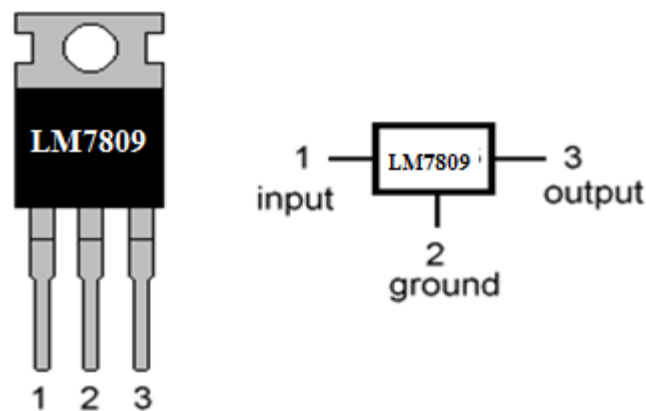


Fig: 4.16 Pin details of IC 7809

IC 7809 is a three terminal linear voltage regulator IC with a fixed output voltage of 9V which is useful in a wide range of applications.

7809 IC Rating:

- Input voltage range 10V- 35V
- Current rating $I_c = 1A$
- Output voltage range $V_{Max}=5.2V$, $V_{Min}=4.8V$

CHAPTER 5

SOFTWARE DETAILS

5.1 EMBEDDED ‘C’

Embedded C is a set of language extensions for the C Programming language by the C Standards committee to address commonality issues that exist between C extensions for different embedded systems. Historically, embedded C programming requires nonstandard extensions to the C language in order to support exotic features such as fixed-point arithmetic, multiple distinct memory banks, and basic I/O operations.

Embedded Systems Programming

- Embedded systems programming is different from developing applications on a desktop computers. Key characteristics of an embedded system, when compared to PCs, are as follows:
- Embedded devices have resource constraints (limited ROM, limited RAM, and limited stack space, less processing power).
- Components used in embedded system and PCs are different; embedded systems typically uses smaller, less power consuming components.
- Embedded systems are more tied to the hardware.

Two salient **features of Embedded Programming** are code speed and code size. Code speed is governed by the processing power, timing constraints, whereas code size is governed by available program memory and use of programming language. Goal of embedded system programming is to get maximum features in minimum space and

minimum time.

Embedded systems are programmed using different type of languages:

- Machine Code
- Low level language, i.e., assembly
- High level language like C, C++, Java, Ada, etc.
- Application level language like Visual Basic, scripts, Access, etc.

Assembly language maps mnemonic words with the binary machine codes that the processor uses to code the instructions. Assembly language seems to be an obvious choice for programming embedded devices. However, use of assembly language is restricted to developing efficient codes in terms of size and speed. Also, assembly codes lead to higher software development costs and code portability is not there. Developing small codes are not much of a problem, but large programs/projects become increasingly difficult to manage in assembly language. Finding good assembly programmers has also become difficult nowadays. Hence high level languages are preferred for embedded systems programming.

Use of **C in embedded systems** is driven by following advantages

- It is small and reasonably simpler to learn, understand, program and debug.
- C Compilers are available for almost all embedded devices in use today, and there is a large pool of experienced C programmers.
- Unlike assembly, C has advantage of processor-independence and is not specific to any particular microprocessor/ microcontroller or any system. This makes it convenient for a user to develop programs that can run on most of the systems.
- As C combines functionality of assembly language and features of high level languages, C is treated as a ‘middle-level computer language’ or ‘high level assembly language’
- It is fairly efficient
- It supports access to I/O and provides ease of management of large embedded projects.

Many of these advantages are offered by other languages also, but what sets C apart from others like Pascal, FORTRAN, etc. is the fact that it is a middle level language; it provides direct hardware control without sacrificing benefits of high level languages.

Compared to other high level languages, C offers more flexibility because C is relatively small, structured language; it supports low-level bit-wise data manipulation.

5.2 PHP

PHP is a server-side scripting language designed for web development but also used as a general-purpose programming language. Originally created by RasmusLerdorf in 1994, the PHP reference implementation is now produced by The PHP Group. PHP originally stood for Personal Home Page, but it now stands for the recursive backronym PHP: Hypertext Preprocessor. PHP code may be embedded into HTML code, or it can be used in combination with various web template systems, web content management system and web frameworks. PHP code is usually processed by a PHP interpreter implemented as a module in the web server or as a Common Gateway Interface (CGI) executable. The web server combines the results of the interpreted and executed PHP code, which may be any type of data, including images, with the generated web page. PHP code may also be executed with a command-line interface (CLI) and can be used to implement standalone graphical applications.

CHAPTER 6 CONCLUSION

The system has been successfully designed and prototyped to measure the human health parameters accurately and it can be monitored through anywhere by using CLOUD and emergency alert message send to admin or supervisor via SMS by using GSM network. The design of the embedded system is very simple and easy to install cum portable hand held device. The obtained results on real group of patients confirmed how the ARDUINO BASED Health framework overcomes the traditional methods in the relationship between doctor and patient making easier the interaction, increasing the number of patients that can be assisted and drastically reducing the dropout rate of patients leaving the pilot.

REFERENCE

- Narmeen zakaria bawany , Tehreem Qamar, Hira Tariq , and Saifullah Adnan “Integrating Healthcare Services Using Blockchain-Based Telehealth Framework” Digital Object Identifier 10.1109/ACCESS.2022.3161944IEEE 2022
- Martin Jürgens*, Dennis Meis Bluetooth Mesh Networks for Indoor Localization International Conference on Mobile Data Management (MDM) IEEE 2019
- Busra Ozlu¹, Bong Sup Shim Flexible and Wearable PEDOT-Paper Pressure Sensor for Detecting Human Voice IEEE -2021 international conference on Big Data and Smart Computing
- Mahmood Khayatzaheh, *Student Member, IEEE* 0.7-V 17.4- μ W 3-Lead Wireless ECG SoC IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS 2013 IEEE
- Prema Sundaram¹ ¹Biomedical Engineering & P.S.N.A College of Engineering and Technology, India Patient monitoring system using android technology IJCSMC, Vol. 2, Issue. 5, May 2013
- Chin-Teng Lin, Fellow, IEEE, Kuan-Cheng Chang, Chun-Ling Lin An Intelligent Telecardiology System Using a Wearable and Wireless ECG to Detect Atrial Fibrillation IEEE TRANSACTIONS ON INFORMATION TECHNOLOGY IN BIOMEDICINE, VOL. 14, NO. 3, MAY 2010
- Joan Gómez-Clapers, Ramon Casanella Wireless ECG and Heart Rate Monitoring Using Dual Ground Dry Electrodes Instrumentation for the ICT Era Sept. 8-10, 2010, Kosice, Slovakia Castelldefels (Barcelona) – SPAIN

APPENDIX

A.SOURCE CODING - ARDUINO

```
#include <SoftwareSerial.h>
#define SIM800_TX_PIN 2
//ARDUINO DIGITAL PIN 3
#define SIM800_RX_PIN 3
//ARDUINO DIGITAL PIN 2
SoftwareSerial serialSIM800(SIM800_TX_PIN,SIM800_RX_PIN);
//SoftwareSerial softserial(12,13);
int sensorPin1 = A0;
int sensorPin2 = A1;
int sensorPin3 = A2;

float tem,hr,bp;
int bzt=10;
int l1=4;
int l2=5;
int l3=6;

void setup()
{
  Serial.begin(9600); // Initialize serial communications with the PC
  while(!Serial);
  //Being serial communication witj Arduino and SIM800
  serialSIM800.begin(9600);
  delay(1000);
  Serial.println("Setup Complete!");
  Serial.println("Sending SMS...");
  //Set SMS format to ASCII
  serialSIM800.write("AT+CMGF=1\r\n");
  delay(1000);
  //Send new SMS command and message number
  serialSIM800.write("AT+CMGS=\"6379995431\"\r\n");
  delay(1000);

  //Send SMS content
  serialSIM800.write("WELCOME TO ACE");
  delay(1000);

  //Send Ctrl+Z / ESC to denote SMS message is complete
```

```

serialSIM800.write((char)26);
delay(1000);

Serial.println("SMS Sent!");

pinMode(bzr, OUTPUT);
pinMode(l1, OUTPUT);
pinMode(l2, OUTPUT);
pinMode(l3, OUTPUT);
digitalWrite(bzr, LOW);
}

void loop() {

    float temp_val,hr1,bp1;
    /* Read Temperature */

    hr = analogRead(sensorPin1);
    delay(50);
    bp=analogRead(sensorPin2);
    delay(50);
    tem = analogRead(sensorPin3);
    delay(50);
    temp_val = (tem * 0.4883)/10; /* Convert adc value to equivalent voltage */
    delay(50);
    hr1=(hr+70);
    bp1=(bp/4);
    delay(100);
    Serial.println ("HEART RATE");
    Serial.println (hr1);
    Serial.println ("BP RATE");
    Serial.println (bp1);
    Serial.println ("TEMP:");
    Serial.println (temp_val);
    delay(500);

    if((hr1>100)&&(hr1<110))
    {
        digitalWrite(bzr, HIGH);
        digitalWrite(l1, HIGH);
        digitalWrite(l2, LOW);
    }
}

```



```

    digitalWrite(l3, LOW);

delay(200);
Serial.println("Sending SMS...");
    serialSIM800.write("AT+CMGF=1\r\n");
    delay(1000);
    serialSIM800.write("AT+CMGS=\"6379995431\"\r\n");
    delay(1000);
    serialSIM800.write("ABNORMAL HEART RATE");
    delay(1000);
    serialSIM800.write((char)26);
    delay(1000);
    Serial.println("SMS Sent!");
    delay(500);
    }
else if((bp1>120)&& (bp1<200))
{
    digitalWrite(bzr, HIGH);
    digitalWrite(l1, LOW);
    digitalWrite(l2, HIGH);
    digitalWrite(l3, LOW);
delay(200);
Serial.println("Sending SMS...");
    serialSIM800.write("AT+CMGF=1\r\n");
    delay(1000);
    serialSIM800.write("AT+CMGS=\"6379995431\"\r\n");
    delay(1000);
    serialSIM800.write("ABNORMAL BLOOD PRESSURE");
    delay(1000);
    serialSIM800.write((char)26);
    delay(1000);
    Serial.println("SMS Sent!");
    delay(500);
    }
    else if((hr1>120)|| (bp1>300))
{
    Serial.print("CHECK BIOMETRIC");
    digitalWrite(bzr, HIGH);
    digitalWrite(l1, HIGH);
    digitalWrite(l2, HIGH);
    digitalWrite(l3, HIGH);

```

```

    delay(2000);
    digitalWrite(bzr, LOW);
    delay(200);

    }
    else if(temp_val>38)
    {
        digitalWrite(bzr, HIGH);
        digitalWrite(l1, LOW);
        digitalWrite(l2, LOW);
        digitalWrite(l3, HIGH);
        delay(200);
Serial.println("Sending SMS...");
        serialSIM800.write("AT+CMGF=1\r\n");
        delay(1000);
        serialSIM800.write("AT+CMGS=\"6379995431\"\r\n");
        delay(1000);
        serialSIM800.write("ABNORMAL BODY TEMPERATURE");
        delay(1000);
        serialSIM800.write((char)26);
        delay(1000);
        Serial.println("SMS Sent!");
        }
        else if((hr1>110)&&(hr1<120)&&(bp1>120)&&(bp1<300))
        {
            digitalWrite(bzr, HIGH);
            digitalWrite(l1, HIGH);
            digitalWrite(l2, HIGH);
            digitalWrite(l3, LOW);
            delay(200);
Serial.println("Sending SMS...");
            serialSIM800.write("AT+CMGF=1\r\n");
            delay(1000);
            serialSIM800.write("AT+CMGS=\"6379995431\"\r\n");
            delay(1000);
            serialSIM800.write("ABNORMAL HEART-BP RATE");
            delay(1000);
            serialSIM800.write((char)26);
            delay(1000);
            Serial.println("SMS Sent!");
            delay(500);

```

```

    }
    else if((hr1>110)&&(hr1<120)&&(temp_val>38))
    {
        digitalWrite(bzr, HIGH);
        digitalWrite(l1, HIGH);
        digitalWrite(l2, LOW);
        digitalWrite(l3, HIGH);
    delay(200);
    Serial.println("Sending SMS...");
        serialSIM800.write("AT+CMGF=1\r\n");
    delay(1000);
        serialSIM800.write("AT+CMGS=\"6379995431\"\r\n");
    delay(1000);
        serialSIM800.write("ABNORMAL HEART-TEMP RATE");
    delay(1000);
        serialSIM800.write((char)26);
    delay(1000);
    Serial.println("SMS Sent!");
        delay(500);
    }
    else if((bp1>120)&&(bp1<300)&&(temp_val>38))
    {
        digitalWrite(bzr, HIGH);
        digitalWrite(l1, LOW);
        digitalWrite(l2, HIGH);
        digitalWrite(l3, HIGH);
    delay(200);
    Serial.println("Sending SMS...");
        serialSIM800.write("AT+CMGF=1\r\n");
    delay(1000);
        serialSIM800.write("AT+CMGS=\"6379995431\"\r\n");
    delay(1000);
        serialSIM800.write("ABNORMAL BP-TEMP RATE");
    delay(1000);
        serialSIM800.write((char)26);
    delay(1000);
    Serial.println("SMS Sent!");
        delay(500);
    }

```

```

    else
    {
    digitalWrite(bzr, LOW);
    digitalWrite(l1, LOW);
    digitalWrite(l2, LOW);
    digitalWrite(l3, LOW);
    delay(500);

    }

}

```

B.SOURCE CODING – WEB PAGE

```

#include <ESP8266WiFi.h>
#include <ESP8266WebServer.h>
#include <ESP8266mDNS.h>
#include "user_interface.h" // Needed for timer

// Define time function for periodic read in of digital input
os_timer_t myTimer;
#define TIMER_UPDATE 1000
// #define analogPin A0

const char* ssid    = "ACEPROJECT";
const char* password = "ACEPROJECT";
String S, msg, msg1, msg2, msg3, msg4;
//const int buzzerPin = 2;
int c1;
int L1 = 16;
int L2 = 5;
int L3 = 4;
// Webserver parameter
ESP8266WebServer server(80);
String webpage_head, webpage_trail, webpage_trail1, webpage_head1, webpage_head2,
webpage_head3, webpage_head4, webpage_head5, webpage_head6, webpage_head7,
webpage_head8;
//String S1, S2;
// Start MDNS
// If MDNS is supported by client, the ESP8266 can be accessed via
// address "http://[mdns_name].local"

```

```

MDNSResponder mdns;
const char* mdns_name = "esp8266";

// Setup routine

void setup() {
  Serial.begin(9600);    // Configure serial port
  delay(10);
  pinMode(L1, INPUT);
  pinMode(L2, INPUT);
  pinMode(L3, INPUT);
  connect_wifi();        // Connect to WiFi

  // Activate MDNS responder
  if (mdns.begin(mdns_name, WiFi.localIP())) {
    Serial.println("MDNS responder started");
  }

  start_httpd();         // Start webserver

  // Configure timer
  os_timer_setfn(&myTimer, timerCallback, NULL);
  os_timer_arm(&myTimer, TIMER_UPDATE, true);
}

// the loop function runs over and over again forever
void loop() {

  //  Serial.print(sub);
  if(digitalRead(L1)==LOW && digitalRead(L2)==LOW && digitalRead(L3)==LOW)
  {
    //Serial.print(S);
    Serial.println("matched");
    c1=1;
  }

  else if (digitalRead(L1)==HIGH && digitalRead(L2)==LOW &&
digitalRead(L3)==LOW)
{
  c1=2;

```

```

}
else if (digitalRead(L1)==LOW && digitalRead(L2)==HIGH &&
digitalRead(L3)==LOW)
{
    c1=3;
}
else if (digitalRead(L1)==LOW && digitalRead(L2)==LOW &&
digitalRead(L3)==HIGH)
{
    c1=4;
}

else if (digitalRead(L1)==HIGH && digitalRead(L2)==HIGH &&
digitalRead(L3)==LOW)
{
    c1=5;
}
else if (digitalRead(L1)==HIGH && digitalRead(L2)==LOW &&
digitalRead(L3)==HIGH)
{
    c1=6;
}
else if (digitalRead(L1)==LOW && digitalRead(L2)==HIGH &&
digitalRead(L3)==HIGH)
{
    c1=7;
}
else if (digitalRead(L1)==HIGH && digitalRead(L2)==HIGH &&
digitalRead(L3)==HIGH)
{
    c1=8;
}
server.handleClient();
}

// Timer-controlled read in of digital input
void timerCallback(void *pArg) {

}

// Conecting to WiFi network

```

```

void connect_wifi() {
  Serial.println();
  Serial.println();
  Serial.println();
  Serial.print("Connecting to ");
  Serial.print(ssid);
  Serial.print(" ");

  WiFi.mode(WIFI_STA); // Explicitly set the ESP8266 to be a WiFi-client
  WiFi.begin(ssid, password);

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println();
  Serial.println("WiFi connected");
  Serial.print("IP address: ");
  Serial.println(WiFi.localIP());
  Serial.print("MAC: ");
  Serial.println(WiFi.macAddress());
  Serial.println();
}

// Setup the webserver
void start_httpd() {

  webpage_head+="<head>";
  webpage_head+="

```

```
// webpage_head+="<h1><center>DEPARTMENT OF EEE </center></h1>";
webpage_head+="<h1><center> ADHIYAMAAN COLLEGE OF ENGINEERING - AI
BASED PATIENT HEALTH MONITORING SYSTEM </center></h1>";
```

```
webpage_head+="<form></br>";
```

```
webpage_head1+="<center><b></br><p> PATIENT HEALTH : NORMAL
<p></br>";
webpage_head1+="</p></center>";
```

```
webpage_head2+="<center><b></br><p> PATIENT HEALTH : ABNORMAL
HEART RATE <p></br>";
webpage_head2+="</p></center>";
```

```
webpage_head3+="<center><b></br><p> PATIENT HEALTH : ABNORMAL
BLOOD PRESSURE <p></br>";
webpage_head3+="</p></center>";
```

```
webpage_head4+="<center><b></br><p> PATIENT HEALTH : ABNORMAL BODY
TEMPERATURE <p></br>";
webpage_head4+="</p></center>";
```

```
webpage_head5+="<center><b></br><p> PATIENT HEALTH : ABNORMAL
HEART AND BLOOD PRESSURE RATE <p></br>";
webpage_head5+="</p></center>";
```

```
webpage_head6+="<center><b></br><p> PATIENT HEALTH : ABNORMAL
HEART RATE AND TEMPERATURE <p></br>";
webpage_head6+="</p></center>";
```

```
webpage_head7+="<center><b></br><p> PATIENT HEALTH : ABNORMAL
BLOOD PRESSURE AND TEMPERATURE <p></br>";
webpage_head7+="</p></center>";
```

```
webpage_head8+="<center><b></br><p> CHECK BIOMETRIC : NO SIGNAL
<p></br>";
webpage_head8+="</p></center>";
```

```
webpage_trail+="</form>";
```

```
webpage_trail+="</body></html>";
```

```
server.on("/", []){
Serial.println("HTTP request");
```

```
if(c1==1)
```



```

{
server.send(200, "text/html", webpage_head+webpage_head1+webpage_trail);
}

else if(c1==2)
{
server.send(200, "text/html", webpage_head+webpage_head2+webpage_trail);
}
else if(c1==3)
{
server.send(200, "text/html", webpage_head+webpage_head3+webpage_trail);
}
else if(c1==4)
{
server.send(200, "text/html", webpage_head+webpage_head4+webpage_trail);
}
else if(c1==5)
{
server.send(200, "text/html", webpage_head+webpage_head5+webpage_trail);
}
else if(c1==6)
{
server.send(200, "text/html", webpage_head+webpage_head6+webpage_trail);
}
else if(c1==7)
{
server.send(200, "text/html", webpage_head+webpage_head7+webpage_trail);
}
else if(c1==8)
{
server.send(200, "text/html", webpage_head+webpage_head8+webpage_trail);
}

}

);

server.begin();
Serial.println("HTTP server started");
} server.begin();
Serial.println("HTTP server started");
}

```